

Trends in nutritional intake and serum cholesterol levels over 40 years in Tanushimaru, Japanese men.



BACKGROUND: Rapid socioeconomic development in Japan since the beginning of the Seven Countries Study in 1958 has brought remarkable changes in lifestyle and dietary patterns. We investigated the relationship between time trends in nutrient intake and serum cholesterol levels in a Japanese cohort of the Seven Countries Study, in Tanushimaru, a typical farming town on Kyushu Island. METHODS: Subjects totaled 628 in 1958, 539 in 1977, 602 in 1982, 752 in 1989, and 402 in 1999, and all of the subjects were men aged 40-64 years. Eating patterns were evaluated by 24-hour dietary recall from 1958 through 1989, and by a food frequency questionnaire in 1999. We also measured serum cholesterol levels in each health examination. RESULTS: The total daily energy intake decreased from 2837 kcal in 1958 to 2202 kcal in 1999. The carbohydrate intake in percentage of total daily energy intake decreased markedly, from 84% in 1958 to 62% in 1999, in contrast to large increases during this period in protein intake (from 11% to 18%) and fat intake (from 5% to 20%). In proportion to the dramatic change in protein and fat intake, serum cholesterol levels showed large increases (from 152.5mg/dl to 194.2 mg/ dL). CONCLUSIONS: In spite of such big dietary changes toward a westernized diet, the incidence of coronary artery disease in a rural Japanese area remains low. However, careful surveillance is needed in the future because of the remarkably increasing intake of fats, especially saturated fatty acids.